

Python Interview Questions – Topic-wise Preparation Guide

1. General Python & Variables

- What is Python? Is it high-level or low-level?
- What are the main features of Python?
- Why is Python called a dynamically typed language?
- What is an interpreted language?
- What are Python keywords?
- Can we use keywords like if or while as variable names?
- What are identifiers?
- What are the rules for naming identifiers?
- What is the difference between variables and literals?
- What are Python data types?
- What are mutable and immutable data types?
- What is the difference between type() and id()?
- What does the id() function return?
- What does the type() function return?
- What is indentation in Python?
- Why is indentation mandatory in Python?
- What are comments?
- What is the difference between comments and docstrings?
- What are docstrings used for?
- What is dynamic memory allocation in Python?

2. Numbers & Operators

- What are the numeric data types in Python?
- What is an integer in Python?
- What is a float in Python?
- What is a complex number in Python?
- What are arithmetic operators?
- Explain +, -, *, /, %, //, **.
- What is floor division?

- What is modulo operator?
- What is the difference between / and //?
- What are comparison operators?
- Explain ==, !=, >, <, >=, <=.
- What are logical operators?
- Explain and, or, not.
- What are assignment operators?
- Explain +=, -=, *=, /=.
- What is the difference between == and is?
- What are identity operators?
- What are membership operators?
- Explain in and not in.

3. Strings

- What is a string in Python?
- Are Python strings mutable or immutable?
- What does string immutability mean?
- What is string indexing?
- What is string slicing?
- What does s[1:5] do?
- How do you reverse a string using slicing?
- What are common string methods?
- What does .split() do?
- What does .partition() do?
- What does .strip() do?
- What does .lstrip() do?
- What does .rstrip() do?
- How do you convert string to uppercase or lowercase?
- What does .isdigit() do?
- What does .isalpha() do?
- How do you find the index of a character?
- What is the difference between .find() and .index()?
- What is string concatenation?
- What is string formatting?
- What are f-strings?

4. List & Tuple

- What is a list?
- What is a tuple?
- What is the difference between list and tuple?
- Are lists mutable or immutable?
- Are tuples mutable or immutable?
- What is list indexing?
- What is negative indexing?
- What is list slicing?
- How do you add elements to a list?
- What does `.append()` do?
- What does `.insert()` do?
- What does `.extend()` do?
- How do you remove elements from a list?
- What does `.remove()` do?
- What does `.pop()` do?
- What does `del` do?
- How do you sort a list?
- What does `.sort()` do?
- What does `.reverse()` do?
- What is tuple unpacking?
- Can tuples contain mutable objects?

5. Dictionary & Set

- What is a dictionary?
- What are key-value pairs?
- Are dictionaries ordered?
- Are dictionary keys mutable or immutable?
- Why must dictionary keys be immutable?
- How do you access values in a dictionary?
- What is the difference between `dict['key']` and `dict.get('key')`?
- How do you add elements to a dictionary?
- How do you remove elements from a dictionary?
- What does `.keys()` return?
- What does `.values()` return?

- What does `.items()` return?
- What is a set in Python?
- Do sets allow duplicate values?
- Are sets ordered or unordered?
- How do you create a set?
- How do you add elements to a set?
- How do you remove elements from a set?
- What is set union?
- What is set intersection?
- What is set difference?

6. Control Flow (If-Else & Loops)

- What is a conditional statement?
- What is the syntax of an if statement?
- What is if-elif-else?
- What is nested if?
- What is a loop?
- What is a for loop?
- What is a while loop?
- What is the difference between for loop and while loop?
- What is the `range()` function?
- What are the parameters of `range()`?
- What does `break` do?
- What does `continue` do?
- What does `pass` do?
- What is an infinite loop?
- How can infinite loops occur?
- How can infinite loops be prevented?
- What is a nested loop?
- Can we use an else block with a loop?

7. Functions

- What is a function?
- Why are functions used?
- How do you define a function?

- What are parameters?
- What are arguments?
- What is the difference between parameters and arguments?
- What are default arguments?
- What are positional arguments?
- What are keyword arguments?
- What are variable-length arguments?
- What is `*args`?
- What is `**kwargs`?
- What is the return statement?
- What happens if a function does not have a return statement?
- What is recursion?
- What is a lambda function?
- What is the scope of variables?
- What is the difference between local and global variables?

8. Built-in Functions

- What does `len()` do?
- What does `type()` do?
- What does `id()` do?
- What does `min()` do?
- What does `max()` do?
- What does `sum()` do?
- What does `input()` do?
- How do you convert input to integer?
- What does `dir()` do?
- What does `enumerate()` do?
- What does `zip()` do?

9. OOP (Object-Oriented Programming)

- What is Object-Oriented Programming?
- What is a class?
- What is an object?
- What is the difference between class and object?
- What is a constructor?

- What is the `__init__` method?
- What is the `self` keyword?
- What is an instance variable?
- What is a class variable?
- What is encapsulation?
- What is inheritance?
- What are types of inheritance?
- What is polymorphism?
- What is method overriding?
- What is method overloading?
- What is abstraction?
- What is an abstract class?
- What is the `super()` function?
- What is Method Resolution Order (MRO)?

10. Exception Handling

- What is an exception?
- What is an error?
- What is the difference between error and exception?
- Why is exception handling important?
- What is a try block?
- What is an except block?
- Can we have multiple except blocks?
- What is the finally block?
- When does the finally block execute?
- What is the else block in exception handling?
- What are built-in exceptions in Python?
- What is the `raise` keyword?
- What is a custom exception?
- How do you create a user-defined exception?
- What is exception propagation?

11. File Handling

- What is file handling in Python?
- Why is file handling important?

- What is the `open()` function?
- What does `close()` function do?
- What are file modes in Python?
- Explain `r`, `w`, `a`, `x`.
- What are binary modes `rb` and `wb`?
- What does `read()` do?
- What does `readline()` do?
- What does `readlines()` do?
- How do you write data to a file?
- How do you append data to a file?
- What is `with open()`?
- Why is `with open()` recommended?
- What is a file pointer?
- What does `seek()` do?
- What does `tell()` do?
- How do you read CSV files in Python?
- How do you read JSON files in Python?